

KICAD 8 EDA DESIGN GUIDE

Building an Arduino Uno From Scratch

A step-by-step presentation on schematic capture, footprint mapping, PCB layout, and manufacturing prep

PHASE 1

Architecture & Sub-Systems

Understanding the essential functional blocks of the Arduino Uno board

Main Hardware Sub-Systems

1. Microcontroller Core

The heart of the board centered around the **ATmega328P** 8-bit AVR microcontroller running at 16 MHz.

Includes high-precision 16MHz crystal oscillator, load capacitors (22pF), RC power-on reset circuitry, and ICSP header for direct programming.

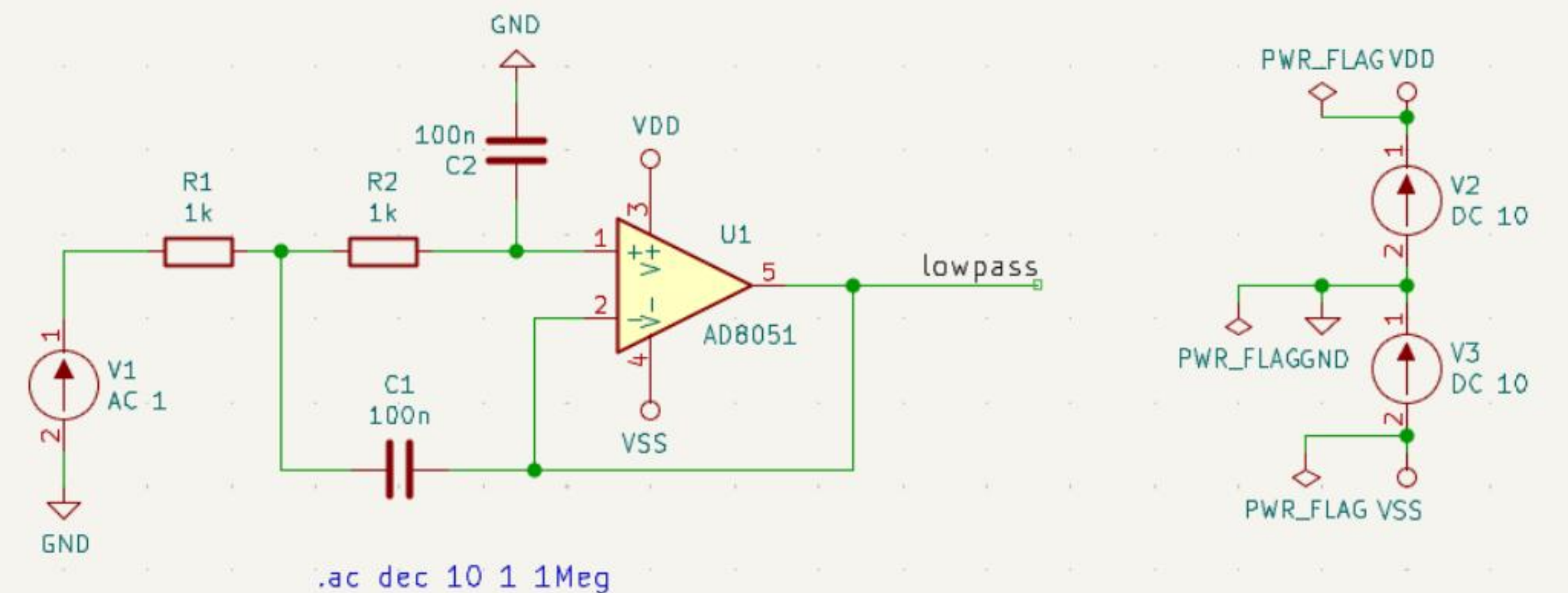
2. USB Interface & Power

Manages external power input and serial communications over USB.

Contains dual LDO regulators (5V & 3.3V), automatic power source switching op-amp circuit, and a USB-to-UART bridge controller (ATmega16U2 or CH340G).

Schematic Capture in Eeschema

- + **Add Symbols:** Press 'A' key to load ATmega328P, passive resistors, capacitors, and connectors.
- 🏷️ **Net Labels:** Use descriptive net labels (TX, RX, RESET, SCK) to keep the schematic clean without long wire lines.
- ⚡ **Power Ports:** Connect explicit power symbols (+5V, +3V3, GND) to ensure clear power distribution.
- 🛡️ **Run ERC:** Execute Electrical Rules Check to detect unconnected pins or output conflicts.



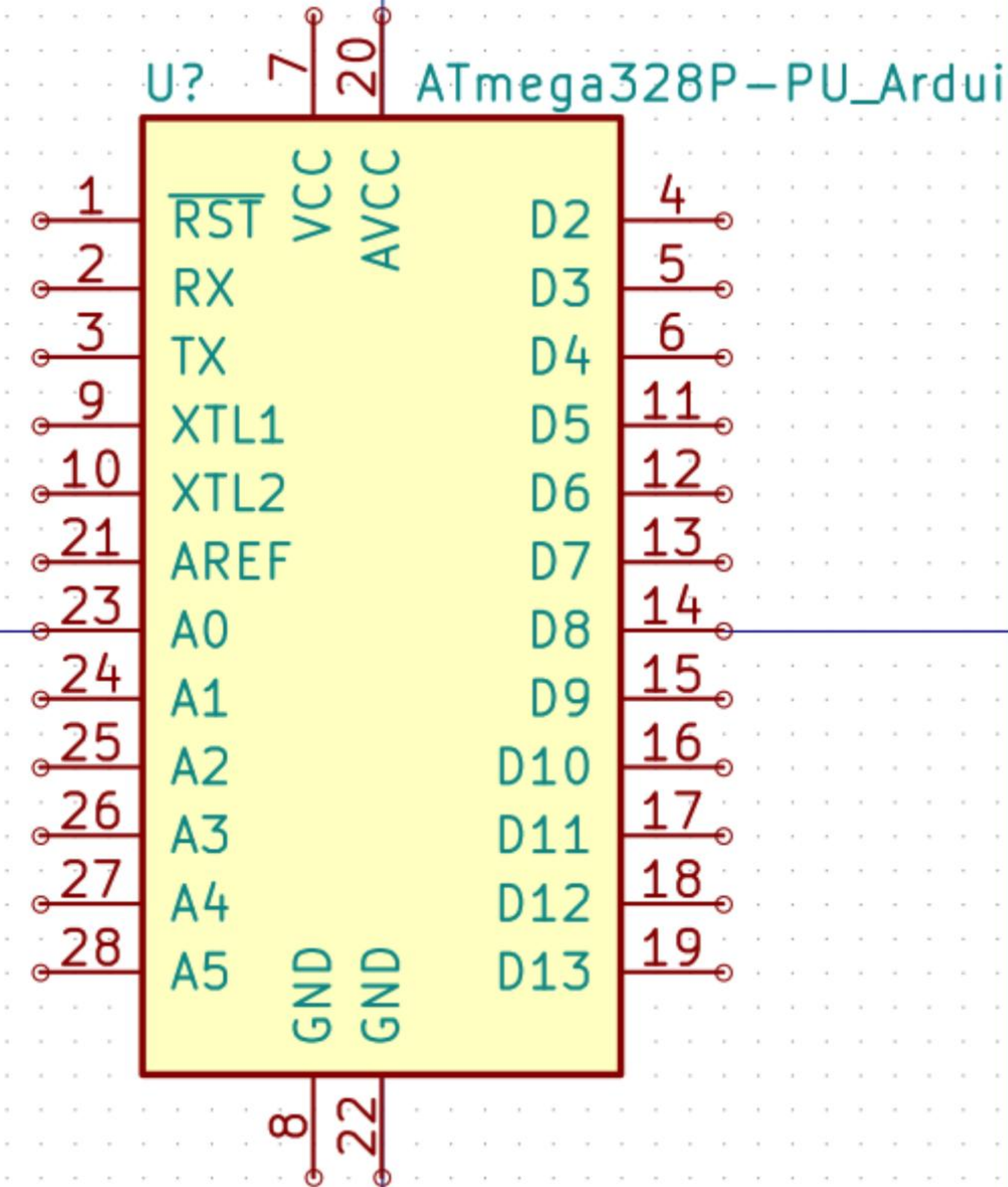
MCU Core Connections

ATmega328P Pinout Setup

Connect crystal across XTAL1 (Pin 9) and XTAL2 (Pin 10) with 22pF caps routed directly to Ground.

Attach a 10k Ω pull-up resistor to the RESET line and a 100nF ceramic capacitor in series with DTR for auto-reset uploading.

Housings_DIP:DIP-28_W7.62mm



Essential Schematic Rules



Decoupling Caps

Place 100nF ceramic decoupling capacitors directly adjacent to every VCC and AVCC pin on the MCU to filter power noise.



16 MHz Crystal

Keep crystal trace connections short and symmetric. Connect symmetrical load capacitors to a clean local ground layer.



Power Filtering

Add an LC filter (10μH inductor + 100nF cap) to the AVCC pin to ensure steady, low-noise analog readings.

Footprint Mapping

100%

Footprint Match Verification

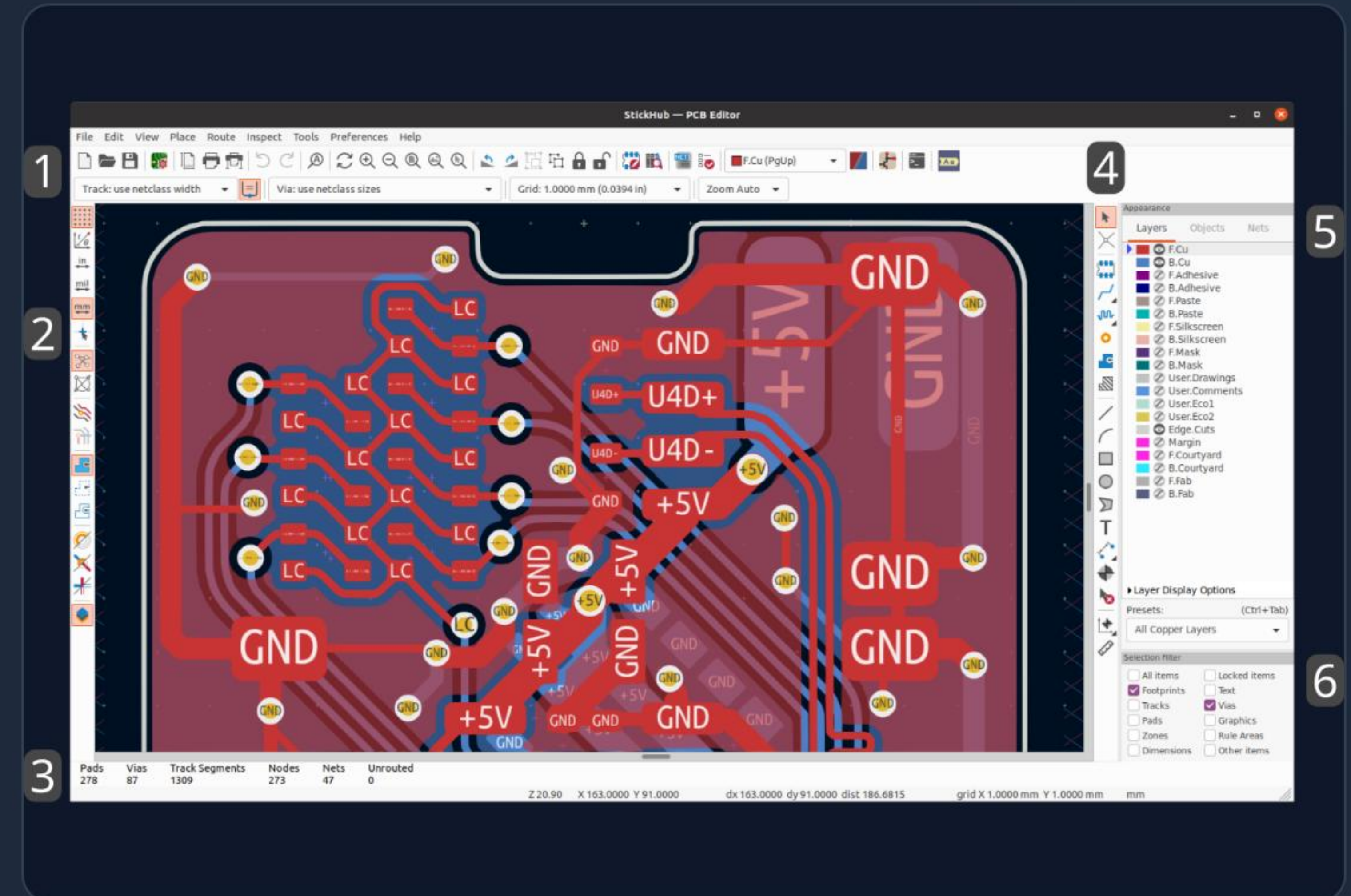
Mapping Symbols to Real Parts

Every schematic symbol must be explicitly paired with a physical footprint (THT DIP-28 or SMD TQFP-32 for the ATmega328P).

Use standard 0603 or 0805 passives for surface mount designs, or standard 2.54mm pitch components for classic through-hole assembly.

PCB Layout & Routing

- ❖ **Component Placement:** Place MCU centrally, position crystal immediately next to XTAL pins.
- 📦 **Ground Plane:** Fill Top (F.Cu) and Bottom (B.Cu) layers with solid GND copper zones.
- 👤 **Trace Sizing:** Route signal lines with 0.25mm width and main power traces with 0.5mm - 1.0mm width.
- 🎯 **Keepout Areas:** Maintain clearance around high-speed crystal and edge connectors.



The Complete PCB Workflow



1. Schematic

Draw circuit blocks and wire nets in Eeschema.



2. Assign Parts

Pair symbols with component PCB footprints.



3. Board Layout

Place parts, route traces, and pour ground planes.



4. Fab Prep

Run DRC check and generate Gerber files.

Bill of Materials (BOM) Key Parts

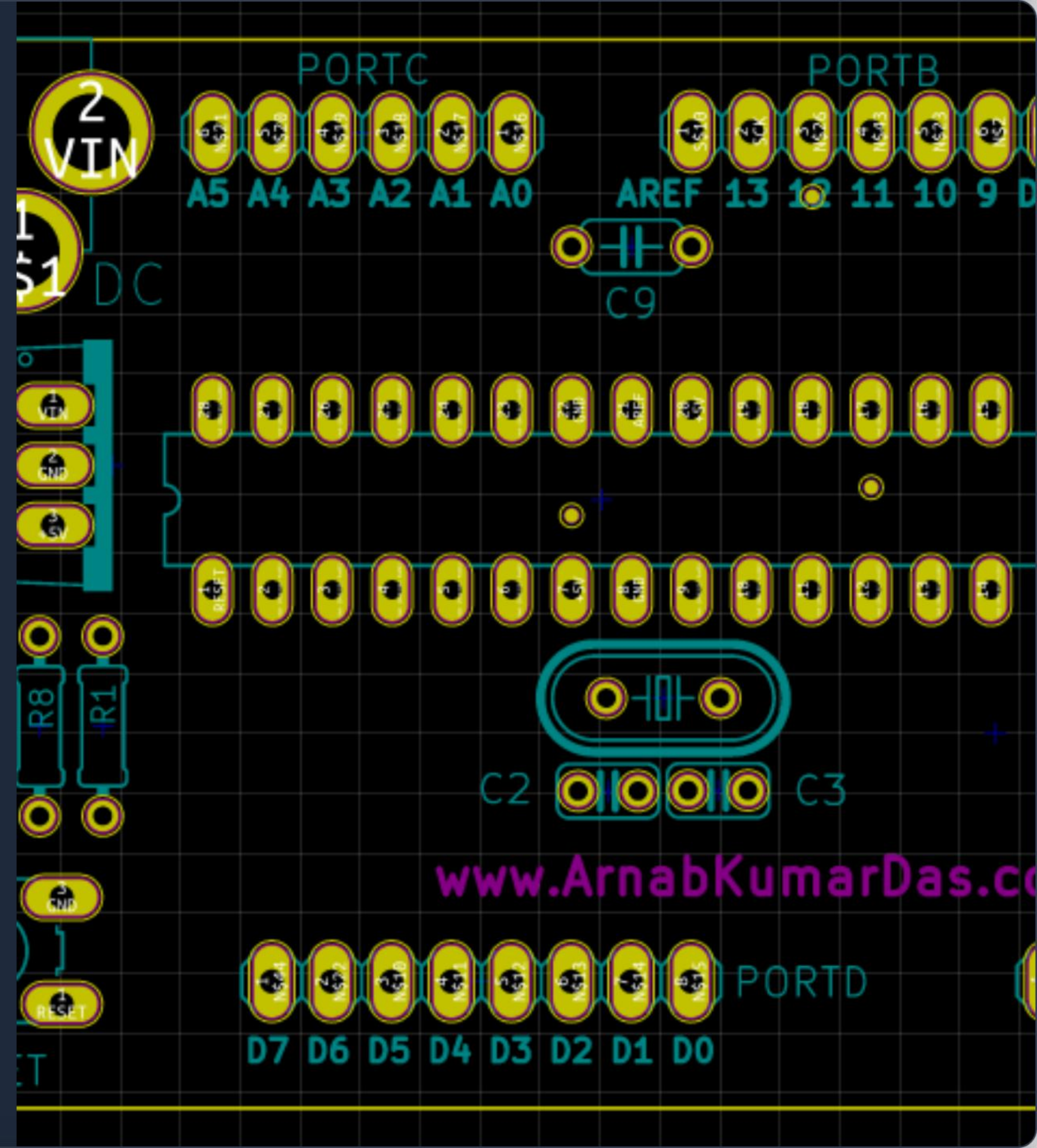
Subsystem	Component Ref	Footprint / Package	Function
MCU Core	ATmega328P-PU	DIP-28 / TQFP-32	Main microcontroller processing chip
Clock	16.000 MHz Crystal	HC-49S / SMD 5032	Main system clock oscillator
Power Rail	NCP1117ST50T3G	SOT-223	5.0V Low Dropout Voltage Regulator
USB Bridge	CH340G / FT232RL	SOIC-16	USB to Serial UART converter

3D Preview & DRC

Final Manufacturing Check

Run **Design Rules Check (DRC)** in KiCad to verify zero clearance violations, open net connections, or overlapping silk screen text.

Open the integrated 3D Viewer (Alt + 3) to physically inspect board geometry, header connector heights, and clearance before sending to fabrication.

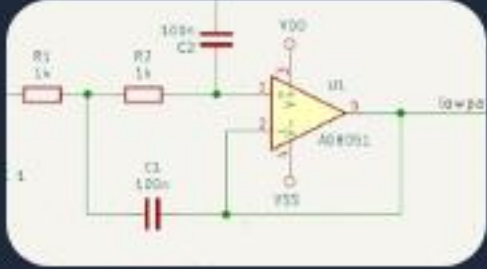


Ready to Build!

Your KiCad Arduino Uno design is ready for fabrication and hand assembly.

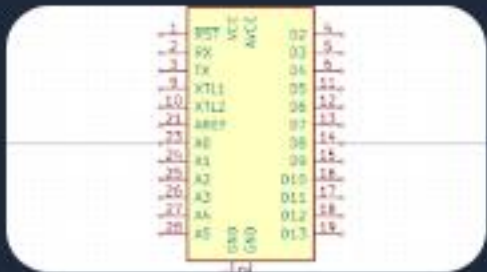
Questions on KiCad setup, schematic capture, or component selection?

Image Sources



https://docs.kicad.org/7.0/en/eeschema/images/eeschema_netlist_schematic.png

Source: docs.kicad.org



<https://user-content.gitlab-static.net/31c79941fdf13bc850fe7a9e001bf4d6dfb1cdc0/68747470733a2f2f757365722d696d616765732e67697468756275736572636f6e74656e742e636f6d2f313038303030382f33313935363833362d39343462353431302d623861312d313165372d383530622d6536343637626130656639322e706e67>

Source: gitlab.com



https://docs.kicad.org/6.0/en/pcbnew/images/pcbnew_user_interface.png

Source: docs.kicad.org



<https://www.arnabkumardas.com/blog/wp-content/uploads/2020/05/DIYArduinoUNO-4.png.webp>

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